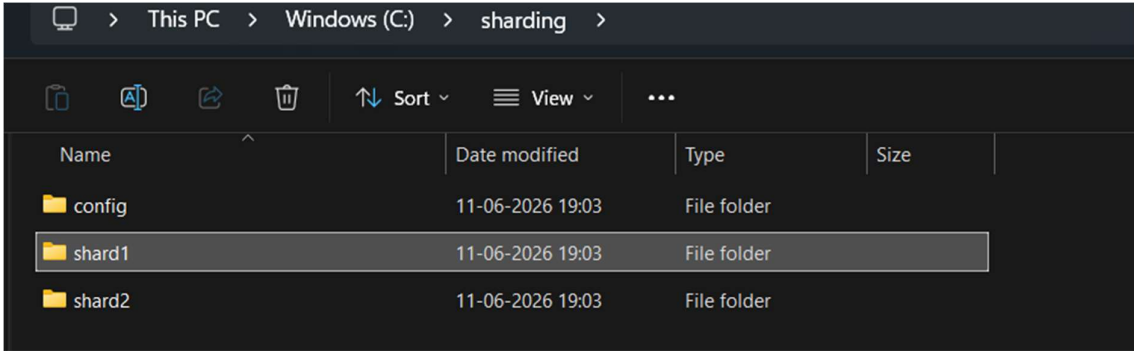


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TOPIC: mongoDB SHARDING



PORT-27019

```
>_MONGOSH

> rs.initiate({
  _id: "configReplSet",
  configsvr: true,
  members: [
    { _id: 0, host: "localhost:27019" }
  ]
})
< {
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1781183611, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1781183611, i: 1 })
}

> rs.status()
< {
  set: 'configReplSet',
  date: 2026-06-11T13:13:41.852Z,
  myState: 1,
  term: Long('1'),
  syncSourceHost: '',
  syncSourceId: -1,
  configsvr: true,
  heartbeatIntervalMillis: Long('2000'),
  majorityVoteCount: 1,
  writeMajorityCount: 1,
  votingMembersCount: 1,
  writableVotingMembersCount: 1,
  optimes: {
    lastCommittedOpTime: { ts: Timestamp({ t: 1781183621, i: 1 }), t: Long('1') },
    lastCommittedWallTime: 2026-06-11T13:13:41.580Z,
    readConcernMajorityOpTime: { ts: Timestamp({ t: 1781183621, i: 1 }), t: Long('1') },
    appliedOpTime: { ts: Timestamp({ t: 1781183621, i: 1 }), t: Long('1') },
    durableOpTime: { ts: Timestamp({ t: 1781183621, i: 1 }), t: Long('1') },
    writtenOpTime: { ts: Timestamp({ t: 1781183621, i: 1 }), t: Long('1') },
    lastAppliedWallTime: 2026-06-11T13:13:41.580Z,
    lastDurableWallTime: 2026-06-11T13:13:41.580Z,
    lastWrittenWallTime: 2026-06-11T13:13:41.580Z
  },
  lastStableRecoveryTimestamp: Timestamp({ t: 1781183611, i: 1 }),
  electionCandidateMetrics: {
    lastElectionReason: 'electionTimeout',
    lastElectionDate: 2026-06-11T13:13:31.324Z,
    electionTerm: Long('1'),
    lastCommittedOpTimeAtElection: { ts: Timestamp({ t: 1781183611, i: 1 }), t: Long('-1') },
    lastSeenWrittenOpTimeAtElection: { ts: Timestamp({ t: 1781183611, i: 1 }), t: Long('-1') },
    lastSeenOpTimeAtElection: { ts: Timestamp({ t: 1781183611, i: 1 }), t: Long('-1') },
    numVotesNeeded: 1,
  }
}
```

```

    lastElectionDate: 2026-06-11T13:13:31.324Z,
    electionTerm: Long('1'),
    lastCommittedOpTimeAtElection: { ts: Timestamp({ t: 1781183611, i: 1 }), t: Long('-1') },
    lastSeenWrittenOpTimeAtElection: { ts: Timestamp({ t: 1781183611, i: 1 }), t: Long('-1') },
    lastSeenOpTimeAtElection: { ts: Timestamp({ t: 1781183611, i: 1 }), t: Long('-1') },
    numVotesNeeded: 1,
    priorityAtElection: 1,
    electionTimeoutMillis: Long('10000'),
    newTermStartDate: 2026-06-11T13:13:31.355Z,
    wMajorityWriteAvailabilityDate: 2026-06-11T13:13:31.491Z
  },
  members: [
    {
      _id: 0,
      name: 'localhost:27019',
      health: 1,
      state: 1,
      stateStr: 'PRIMARY',
      uptime: 72,
      optime: { ts: Timestamp({ t: 1781183621, i: 1 }), t: Long('1') },
      optimeDate: 2026-06-11T13:13:41.000Z,
      optimeWritten: { ts: Timestamp({ t: 1781183621, i: 1 }), t: Long('1') },
      optimeWrittenDate: 2026-06-11T13:13:41.000Z,
      lastAppliedWallTime: 2026-06-11T13:13:41.580Z,
      lastDurableWallTime: 2026-06-11T13:13:41.580Z,
      lastWrittenWallTime: 2026-06-11T13:13:41.580Z,
      syncSourceHost: '',
      syncSourceId: -1,
      infoMessage: 'Could not find member to sync from',
      electionTime: Timestamp({ t: 1781183611, i: 2 }),
      electionDate: 2026-06-11T13:13:31.000Z,
      configVersion: 1,
      configTerm: 1,
      self: true,
      lastHeartbeatMessage: ''
    }
  ],
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1781183621, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1781183621, i: 1 })
}
configReplSet [direct: primary] test>

```

PORT-27011

```

>_MONGOSH                                mongosh: 127.0.0.1:27011
> rs.initiate({
  _id: "shard1ReplSet",
  members: [
    { _id: 0, host: "localhost:27011" }
  ]
})
< {
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1781183776, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1781183776, i: 1 })
}
shard1ReplSet [direct: primary] test>

```

PORT-27012

```

>_MONGOSH                                mongosh: 127.0.0.1:27012
> rs.initiate({
  _id: "shard2ReplSet",
  members: [
    { _id: 0, host: "localhost:27012" }
  ]
})
< {
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1781183845, i: 1 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1781183845, i: 1 })
}
shard2ReplSet [direct: primary] test>

```

PORT-27017

```
> _MONGOSH 2.5.0.1-27087
```

```
> sh.addShard("shard1ReplSet/localhost:27011")
```

```
< {
  shardAdded: 'shard1ReplSet',
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1781183926, i: 58 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1781183926, i: 58 })
}
```

```
> sh.addShard("shard2ReplSet/localhost:27012")
```

```
< {
  shardAdded: 'shard2ReplSet',
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1781183940, i: 11 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1781183940, i: 11 })
}
```

```
> sh.status()
```

```
<
  shardingVersion
  {
    _id: ObjectId('6a2ab47b08af89fbd4022fc7'),
    clusterId: ObjectId('6a2ab47b08af89fbd4022fc6')
  }

```

shards

```
[
  {
    _id: 'shard1ReplSet',
    host: 'shard1ReplSet/localhost:27011',
    topologyTime: Timestamp({ t: 1781183926, i: 32 }),
    replSetConfigVersion: -1
  },
  {
    _id: 'shard2ReplSet',
    host: 'shard2ReplSet/localhost:27012',
    topologyTime: Timestamp({ t: 1781183939, i: 33 }),
    replSetConfigVersion: -1
  }
]
```

mongosh: 127.0.0.1:27017

active mongoses

```
[
  {
    '8.3.2': 1
  }
]
```

autosplit

```
{
  'Currently enabled': 'yes'
}
```

balancer

```
{
  'Currently enabled': 'yes',
  'Currently running': 'no',
  'Failed balancer rounds in last 5 attempts': 0,
  'Migration Results for the last 24 hours': 'No recent migrations'
}
```

shardedDataDistribution

```
[]
```

databases

```
[
  {
    database: {
      _id: 'config',
      primary: 'config',
      partitioned: true
    },
    collections: {}
  }
]
> sh.enableSharding("eCommerceDB")
< {
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1781183964, i: 22 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1781183964, i: 22 })
}
```

mongosh:127.0.0.1:27017

```
> sh.shardCollection(
  "eCommerceDB.orders",
  { customer_id: "hashed" }
)
< {
  collectionssharded: 'eCommerceDB.orders',
  ok: 1,
  '$clusterTime': {
    clusterTime: Timestamp({ t: 1781183976, i: 51 }),
    signature: {
      hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
      keyId: Long('0')
    }
  },
  operationTime: Timestamp({ t: 1781183976, i: 51 })
}
> use eCommerceDB
for(let i=0;i<1000;i++){
  db.orders.insertOne({
    customer_id:"cust_"+i,
    amount:Math.random()*100
  })
}
< switched to db eCommerceDB
> db.orders.getShardDistribution()
<
Shard shard1ReplSet at shard1ReplSet/localhost:27011

{
  data: '0B',
  docs: 0,
  chunks: 1,
  'estimated data per chunk': '0B',
  'estimated docs per chunk': 0
}

Shard shard2ReplSet at shard2ReplSet/localhost:27012

{
  data: '0B',
  docs: 0,
  chunks: 1,
  'estimated data per chunk': '0B',
  'estimated docs per chunk': 0
}
```

```

mongosh:127.0.0.1:27017 > use ecommerce; on shard'
    ]
  }
}
> for(let i=0;i<1000;i++){
  db.orders.insertOne({
    customer_id:"cust_"+i,
    amount:Math.random()*100
  })
}
< {
  acknowledged: true,
  insertedId: ObjectId('6a2ab672bb68193e744983ac')
}
> db.orders.getShardDistribution()
<
Shard shard2ReplSet at shard2ReplSet/localhost:27012

{
  data: '31KiB',
  docs: 509,
  chunks: 1,
  'estimated data per chunk': '31KiB',
  'estimated docs per chunk': 509
}

Shard shard1ReplSet at shard1ReplSet/localhost:27011

{
  data: '30KiB',
  docs: 491,
  chunks: 1,
  'estimated data per chunk': '30KiB',
  'estimated docs per chunk': 491
}

Totals

{
  data: '62KiB',
  docs: 1000,
  chunks: 2,
  'Shard shard2ReplSet': [
    '50.9 % data',
    '50.9 % docs in cluster',
    '63B avg obj size on shard'
  ],
  'Shard shard1ReplSet': [
    '49.09 % data',
    '49.1 % docs in cluster',
    '63B avg obj size on shard'
  ]
}
[mongos] eCommerceDB>

```

